NETWORK CABLE TYPES AND TOOLSTHEORY

1.List 3 key differences between Ethernet cables and fiber optic cables, and give one real-world example where each type would be used (for example, home WiFi router, data center, college campus backbone).

Ans.

Ethernet cables and fiber optic cables are both used to connect devices to a network. However, they differ in the way they transmit data, their speed, and their transmission distance.

No.	Ethernet Cable	Fiber Optic Cable
1. Data Transmission	It transmits data using electrical signals.	It transmits data using light signals.
2. Speed	It provides good speed for home and office networks.	It supports very high speeds and large amounts of data.
3. Distance	It is commonly used for connections up to 100 metres per Ethernet link.	It can transmit data over much longer distances with low signal loss.

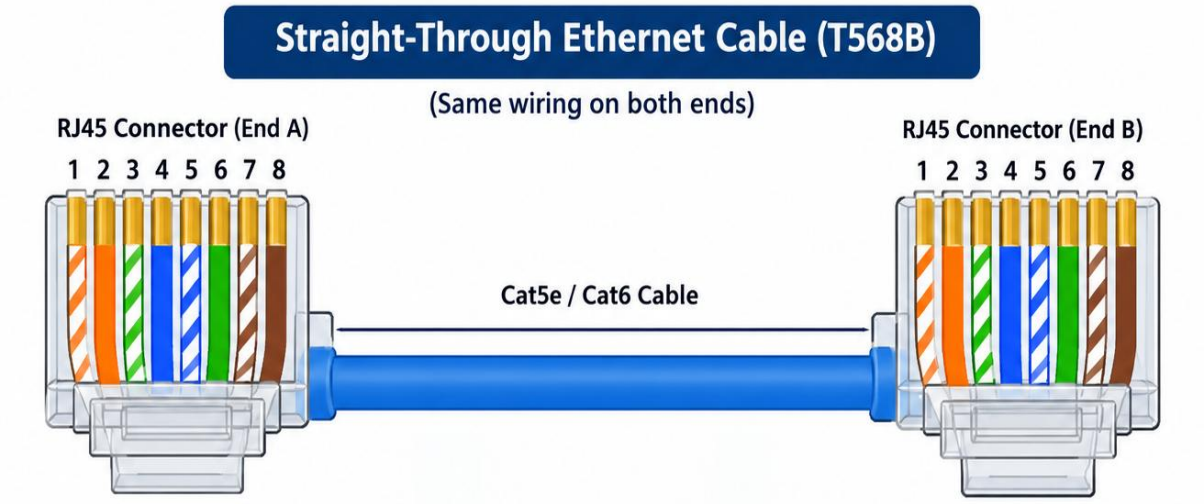
Real-World Examples

1. Ethernet Cable: It is used to connect a computer or gaming PC to a home Wi-Fi router or network switch for a stable internet connection.

2. Fiber Optic Cable: It is used in college campus networks and data centres to connect different buildings or network equipment over long distances at high speed.

2.Draw and label a diagram showing the internal wiring of a straight-through Ethernet cable (use T568B wiring standard), indicating the color of each wire and its pin number.

Ans.



T568B Wiring Standard

Pin No.	Color	Wire Name
1	White/Orange	TX+ (Transmit +)
2	Orange	TX− (Transmit −)
3	White/Green	RX+ (Receive +)
4	Blue	(Not used in 10/100, used for PoE/other)
5	White/Blue	(Not used in 10/100, used for PoE/other)
6	Green	RX− (Receive −)
7	White/Brown	(Not used in 10/100, used for PoE/other)
8	Brown	(Not used in 10/100, used for PoE/other)

Key Points:

- This is a straight-through cable, so both ends use the same T568B wiring standard.
- The wire order is identical on both RJ45 connectors (End A and End B).
- Each pair is twisted together to reduce interference and improve signal quality.

T568B Wire Order (Left to Right in RJ45):

1. White/Orange
2. Orange
3. White/Green
4. Blue
5. White/Blue
6. Green
7. White/Brown
8. Brown

3.Watch a YouTube video demonstrating how to use an RJ45 crimper, then write a step-by-step guide (in your own words) on how to attach an RJ45 connector to a CAT6 cable.

Ans.

Tools and materials required

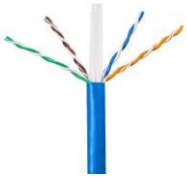
- **CAT6 Ethernet cable**
- **RJ45 connector compatible with CAT6 cable**
- **RJ45 crimping tool**
- **Cable cutter or wire stripper**
- **LAN cable tester**

Procedure



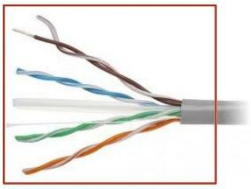
Step 1: Cut the cable

Cut the CAT6 cable to the required length using a cable cutter. Keep a little extra length to make adjustments if needed.



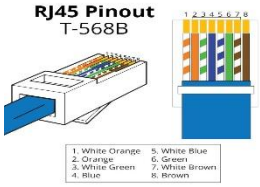
Step 2: Remove the outer covering

Use the stripping tool on the crimper to remove about 2–3 cm of the outer jacket. Be careful not to cut or damage the inner wires.



Step 3: Separate and straighten the wires

Gently untwist the four pairs and straighten all eight wires with your fingers. Avoid untwisting more wire than necessary.



Step 4: Arrange the wires in T568B order

Place the eight wires in the following order, from pin 1 to pin 8:

1. **White/Orange**
2. **Orange**
3. **White/Green**
4. **Blue**
5. **White/Blue**
6. **Green**
7. **White/Brown**
8. **Brown**



Step 5: Trim and insert the wires

Hold the wires firmly in the correct order and cut their ends evenly. Insert them into the RJ45 connector, making sure every wire reaches the front of its channel and the cable jacket extends inside the connector.



Step 6: Crimp the connector

Place the RJ45 connector into the correct slot of the crimping tool. Squeeze the handles firmly so the metal contacts connect with the wires and the connector grips the cable jacket.



Step 7: Test the cable

Remove the connector from the crimper and check that it is fitted properly. Use a LAN cable tester to verify that all eight pins are connected in the correct order.

4.Imagine you are setting up a gaming café with 10 PCs for LAN gaming. Decide which cable type (Ethernet or fiber optics) and which tool (RJ45 crimper or fiber cleaver) you would use, and explain your choices in 3-4 sentences

Ans.

1. Cable Type: I would use CAT6 Ethernet cables to connect all 10 PCs because they provide high speed and a stable connection for LAN gaming.

2. Network Connection: All 10 PCs would be connected to a central network switch using Ethernet cables.

3. Tool Used: I would use an RJ45 crimping tool to attach RJ45 connectors to the CAT6 cables.

4. Reason: Ethernet cables are affordable and easy to install for short distances, while fiber optic cables are more suitable for long-distance connections.